

Persistence will help you finish a CodeWars program.

Mathematical Persistence is the count of how many times you can multiply the digits of a number until you finish in a single digit.

For example, consider the number 7117.



1: Multiply $7 * 1 * 1 * 7$ to get 49.

2: Multiply $4 * 9$ to get 36.

3: Multiply $3 * 6$ to get 18.

4: Multiply $1 * 8$ to get 8.

So, 7117 has a Persistence of 4: $7117 \rightarrow 49 \rightarrow 36 \rightarrow 18 \rightarrow 8$.

Only single digits (0-9) have a Persistence of 0 (no multiplication needed.)

Between 0 and 100 million, the largest Persistence is 9 (only 7448 numbers reach 9.) And over half have a Persistence of 1 because many include 0 as a digit. Here is a table of Persistence and the first number with that Persistence. You will be searching for Persistence.

Persistence:	0	1	2	3	4	5	6	7	8	9
First:	0	10	25	39	77	679	6788	68889	2677889	26888999

Input

Each line of input holds two integers and a letter ("N P X"), separated by single spaces.

The first integer (N, from 1 to 99999999) is the number to start your search.

The second integer (P, from 0 to 9) is the Persistence you're searching for.

The letter (X) can be:

"N" = Ignore P. Simply find the Persistence of N.
 "G" = Search for the first number (N or Greater) that has a Persistence of P. Search up to 100,000,000.
 "L" = Search for the first number (N or Lower) that has a Persistence of P. Search down to 0.
 "S" = Stop. The input lines end with "0 0 S".

Example

```
99999999 0 L
77 0 N
31000000 8 G
76 4 L
99999992 5 G
12345 9 G
0 0 S
```

Output

Reprint each original line, a colon and space, then your result.

Either:

- "Persistence P does not exist between N and (0 or 100000000).", or
- "Z has a Persistence of P."

For any $Z > 9$ also print on a new line four spaces followed by a summary of the multiplication values from Z down to a single digit as shown (with " -> " between each step.)

```
99999999 0 L: 9 has a Persistence of 0.
77 0 N: 77 has a Persistence of 4.
  77 -> 49 -> 36 -> 18 -> 8
31000000 8 G: 31477889 has a Persistence of 8.
  31477889 -> 338688 -> 27648 -> 2688 -> 768 -> 336 -> 54 -> 20 -> 0
76 4 L: Persistence 4 does not exist between 76 and 0.
99999992 5 G: Persistence 5 does not exist between 99999992 and 100000000.
12345 9 G: 26888999 has a Persistence of 9.
  26888999 -> 4478976 -> 338688 -> 27648 -> 2688 -> 768 -> 336 -> 54 -> 20 -> 0
```

Discussion

Your program must print an answer within 5 seconds for each line of input.

Be smart! Use the information above to skip searches. For example, the last line in the Input above asks for $P=9$ above 12345; but you don't have to calculate millions of options since the table gives it to you.

The datasets are constructed such that needed searches will finish within 1,000,000 of N.

See the student datasets for other examples of searches to skip.